PRODUCT REQUIREMENTS SPECIFICATION (PRD)

Supply Chain Cold-Chain Integrity Advisory

Predictive Route Climate Engineering & Real-Time In-Transit Compliance Platform

VERSION	TARGET DOMAIN	COMPLIANCE STANDARDS	STATUS
1.0 (MVP Scope)	Vaccines, Biopharma, Food	FDA 21 CFR Part 11 / WHO GDP	Ready for Engineering Sign-off

1. Executive Summary & Problem Statement

Context: High-potency biopharmaceuticals (vaccines, biologics, gene therapies) and fresh perishable foods require strict, unbroken temperature control throughout the supply chain. A single thermal breach (excursion) can irreversibly spoil entire batches, causing massive financial write-offs and risking patient safety.

The Problem: Existing solutions rely on passive loggers (where temperature data is only extracted after delivery, when cargo is already ruined) or generic GPS telematics that fail to predict thermal degradation or route climate risks.

> ***The 'Advisory' Advantage:*** *Rather than just alerting after an excursion occurs, our platform provides* ***predictive route risk engineering****. By combining digital route mapping with real-time ambient weather forecasts and vehicle insulation decay models, the platform proactively calculates departure windows, detects transit heatwaves, and routes trucks to certified re-icing and cold-storage facilities before product degradation occurs.*

2. Competitor Benchmarking & Gap Analysis

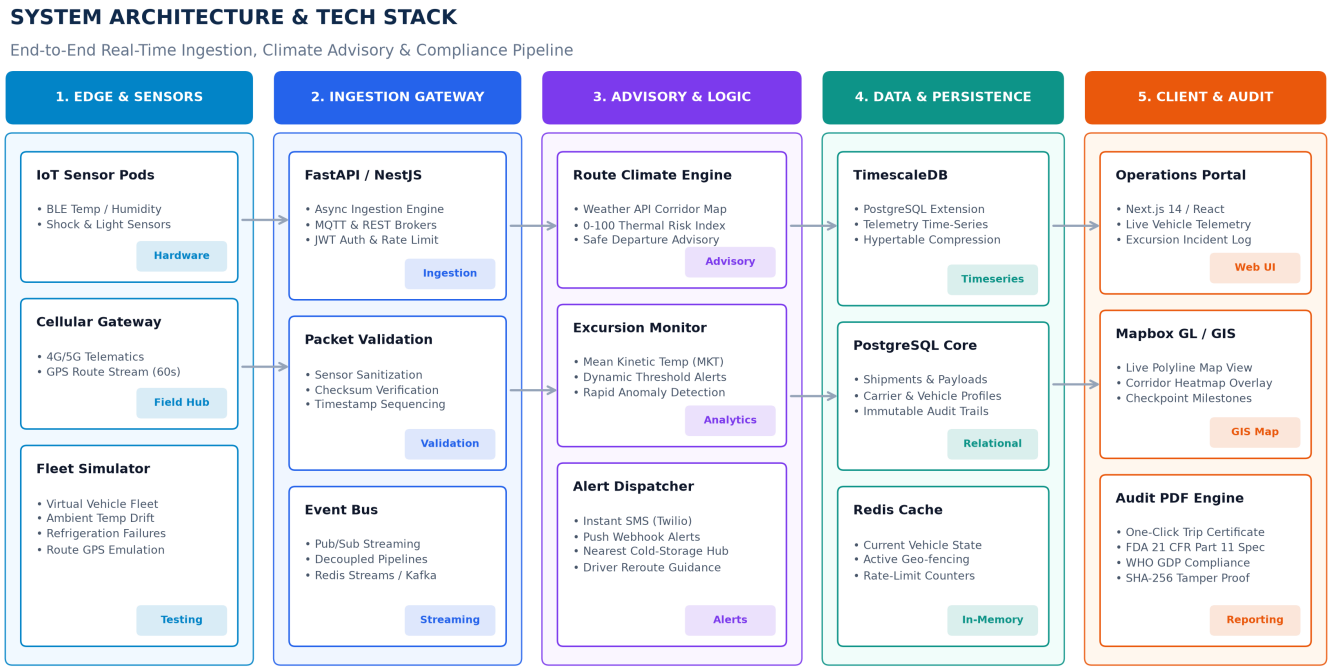
Competitor	Focus Market	Hardware Model	Key Strengths	Market Gaps / Disadvantages
Controlant	Tier-1 Biopharma (Pfizer Vaccine partner)	Proprietary reusable IoT loggers (Saga Card)	Deep pharma validation, automated reverse logistics.	High cost, hardware lock-in, slow onboarding for mid-market distributors.
Roambee	Multimodal Global Supply Chain	BLE beacon swarm + cellular gateway	Condition monitoring, dynamic ETA, item-level tracking.	Complex UI, steep learning curve, requires extensive configuration.
Sensitech (Carrier)	Enterprise Life Sciences & Food	TempTale loggers (USB & Real-time)	Massive global footprint and proven audit history.	Legacy UI/UX, proprietary closed ecosystem, sluggish API integrations.
Tive	Multi-industry freight visibility	Single-use & Multi-use 5G Trackers	Modern web dashboard, multi-sensor (Temp, Humidity, Shock, Light).	Primarily tracking; lacks algorithmic route climate modeling and advisory.

3. Operational Workflow Lifecycle



4. Recommended System Architecture & Tech Stack

The system is built as a microservice-ready cloud architecture designed for high-frequency IoT streaming, algorithmic route calculation, and tamper-proof compliance generation.



Layer	Selected Technology	Architectural Purpose & Rationale
Frontend UI	Next.js 14 / React + Tailwind CSS + Mapbox GL	High-performance operations control tower, live polyline rendering, telemetry charting.
Backend API	FastAPI (Python) or NestJS (Node.js)	Async ingestion of IoT telemetry packets, route advisory math, webhook dispatch.
Database	PostgreSQL with TimescaleDB extension	Time-series hypertable compression for telemetry + relational tables for shipments & audit logs.
In-Memory Cache	Redis 7.x	Real-time geofence indexing and latest-device-state fast lookup.
Field Simulator	Python Fleet Simulator Script	Emulates active truck runs, GPS movements, and dynamic temperature drift for demo/testing.

5. MVP Functional Requirements Scope

Module 1: Pre-Transit Route & Risk Advisory Engine

• **Standard Payload Profiles:** Ultra-Cold (-80°C to -60°C), Standard Cold (2°C to 8°C), Controlled Room Temp (15°C to 25°C), and Frozen Food (-20°C to -10°C).

• **Corridor Weather Forecasting:** Ingests ambient route weather at 50-mile increments to identify ambient heatwaves or sub-zero freezing zones.

• **Advisory Risk Scoring:** Computes a 0–100 risk index factoring transit duration, vehicle cooling rating, and re-icing station proximity.

Module 2: In-Transit Real-Time Telemetry Dashboard

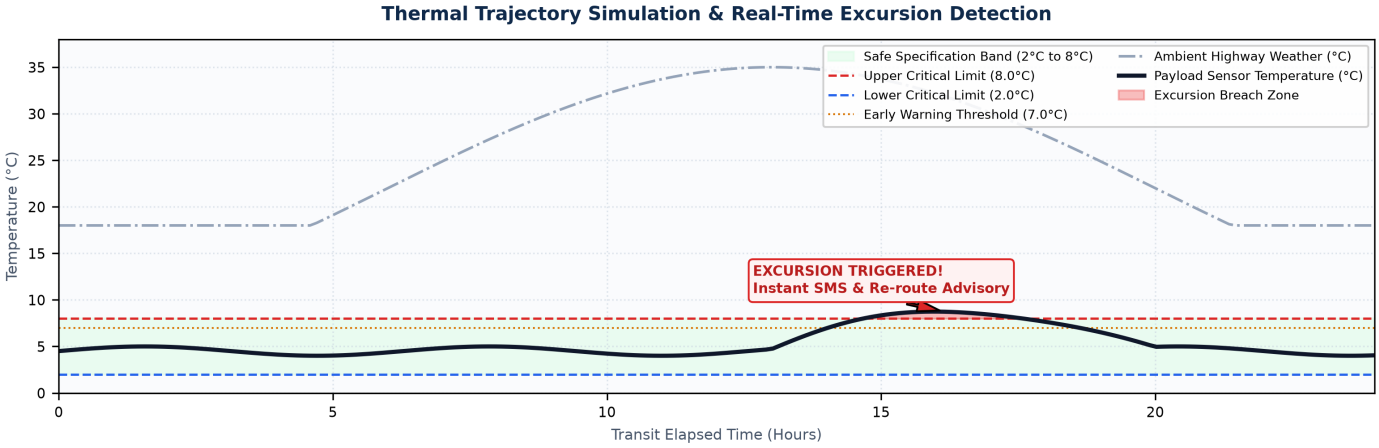
• **Hardware-Agnostic Telemetry API:** Lightweight REST/MQTT endpoint accepting JSON packets (Device ID, Lat/Lng, Temp, Humidity, Battery, Door Sensor).

• **Interactive Transit Map:** Color-coded route path (Green = Safe, Amber = Warning, Red = Excursion) powered by Mapbox GL / Leaflet.

Module 3: Excursion Detection & Mean Kinetic Temperature (MKT)

• **Dynamic Threshold Triggers:** Instant alerting when readings exceed limits for > 15 cumulative minutes.

• **Mean Kinetic Temperature (MKT):** Calculates actual biological degradation via Arrhenius kinetics to evaluate product stability.



Module 4: Regulatory Compliance & Audit Reporting

• **One-Click Trip Audit Certificate:** Automated generation of signed, tamper-evident PDF inspection summaries with temperature curves.

• **Regulatory Conformance:** Built to comply with **FDA 21 CFR Part 11** (audit trail immutability) and **WHO Good Distribution Practices (GDP) Annex 5**.

6. 4-Week MVP Milestone Roadmap

• **Week 1 (Data Foundation):** Relational & time-series schemas, Ingestion API, and realistic mock IoT sensor data generator.

• **Week 2 (Route Advisory & Weather):** Route corridor calculation API, weather forecast integration, and thermal risk scoring.

• **Week 3 (Live Operations UI):** Mapbox visualizer, live sensor graphs via WebSockets, and threshold excursion alerting.

• **Week 4 (Compliance & Hardening):** PDF audit report generator, FDA/WHO compliance validation, end-to-end demo testing.